Week 3 Logic Lab

Propositions and Clear Claims

Purpose: Apply Week 3 skills to new examples. Do not copy examples from the reading handout.

Part 1: Propositions and Structure

Part 1 Q1: "Please submit the form by Friday." Is this a proposition? Explain briefly.

Part 1 Q1 ANSWER BOX

Part 1 Q2: "Some volunteers missed the safety briefing." Identify the subject, predicate, quality, and quantity.

Part 1 Q2 ANSWER BOX

Part 2: Standard Form (A, E, I, O)

Part 2 Q1: "No smartphones in this lab are devices that may be used during the exam." Name the standard form (A, E, I, or O), quality, and quantity.

Part 2 Q1 ANSWER BOX

Part 2 Q2: "At least one finalist is a student who speaks three languages." Name the standard form (A, E, I, or O), quality, and quantity.

Part 2 Q2 ANSWER BOX

Part 3: Translate Ordinary Language

Part 3 Q1: "Only employees with a valid badge may enter the server room." Translate into a clear standard-form categorical proposition. State the subject and predicate you chose.

Part 3 Q1 ANSWER BOX

Part 3 Q2: "Not every ticket holder was seated before the show began." Translate into a clear standard-form categorical proposition. State the subject and predicate you chose.

Part 3 Q2 ANSWER BOX

Part 4: Distribution and Clarity

Part 4 Q1: For the standard-form proposition you wrote in Part 3 Q2, which term is distributed and which is undistributed? Briefly explain.

Part 4 Q1 ANSWER BOX

Part 4 Q2: Read the exchange. Explain what quantity Person A left unclear and how that affects whether Person B's reply refutes the claim.

Person A: "Coaches are unfair to substitutes." Person B: "That is false. My coach gave me extra practice time last week."

Part 4 Q2 ANSWER BOX

Part 5: Review

Part 5 Q1: "The software update fixed every bug reported last month, so the app must be secure against all future attacks." Use the four-step test.

Part 5 Q1 ANSWER BOX

Part 5 Q2: "You have a right to appeal the grade, but that does not mean your answer is right." Which word is used in two different senses? State both meanings.

Part 5 Q2 ANSWER BOX

Part 5 Q3: "All mammals are warm-blooded. Whales are mammals. Therefore whales are warm-blooded." Name the standard form of each statement (A, E, I, or O) and state whether the conclusion follows from the premises.

Part 5 Q3 ANSWER BOX